GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Chemical Engineering

Subject Code: DI05005031

Subject Name: Chemical Reaction Engineering

w. e. f. Academic Year:	2026-27
Semester:	V
Category of the Course:	PEC-III

Prerequisite:	Basic understanding of **Mathematics** (especially calculus and differential equations) to solve rate and reactor design equations. Knowledge of **Physical Chemistry**, particularly chemical kinetics, equilibrium, and thermodynamics. Strong foundation in **Material & Energy Balance** and basic fluid flow concepts in chemical engineering.
Rationale:	To design Chemical reactor, engineer must have knowledge about many areas of chemical engineering like thermodynamics, chemical kinetics, fluid mechanics, heat transfer, mass transfer, and economics. Chemical reaction engineering integrates the aspects which are required for the appropriate design of chemical reactor. For design and operation of the commercial reactors performing different kind of chemical reactions, essential concepts of chemical reaction engineering are required. This course make diploma engineer to perform task of selecting, sizing and determining the optimal operating conditions for the reactor.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Describe fundamental aspects of various chemical reactions.	R, U
02	Determine rate, rate constant, activation energy and order of reaction.	R, U, A
03	Analyze kinetic data to find out rate equation for batch reactor.	R, U, A
04	Control different reactors efficiently using fundamental knowledge of their functioning.	U, A
05	Estimate volume, space time and space velocity for Ideal reactors.	R, U, A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I Basics of Chemical Reactions	1.1 Scope and importance of chemical reaction engineering 1.2 Classification of chemical reactions, a. Homogeneous vs. Heterogeneous, b. Catalytic vs. Non-catalytic c. Reversible vs. Irreversible d. By Molecularity e. Exothermic vs. Endothermic f. By order of reaction 1.3 Introduction to Biochemical reaction 1.4 Reaction rate on various basis and variables affecting the rate of reaction	7	15
Unit– II Kinetics of Homogeneous Reactions	2.1 Rate equation/ Rate law 2.2 Concentration dependent term of rate Equation, Rate constant, Elementary and non-elementary reactions 2.3 Molecularity and order of reaction 2.4 Temperature dependent term of rate Equation, Temperature dependency from Arrhenius law 2.5 Activation energy	8	17
Unit– III Interpretation of batch reactor data	3.1 Methods for analysis of kinetic data Differential vs. Integral method Half-life method 3.2 Relationship for constant volume batch reaction system 3.3 Total pressure of the system and the partial pressure of reacting material Concentration and Conversion 3.4 Integrated rate equation for different order of irreversible reactions: Unimolecular first order, Bi-molecular, Second order, Tri-molecular third order, nth order, Zero order	9	20
Unit– IV Ideal reactors	4.1 Features of ideal reactors 4.2 Different types of reactors: Batch reactor, Semi batch reactor, Flow reactors, MFR/CSTR, PFR (Tubular), Fixed bed reactors, Fluidized bed reactors 4.3 Multi phase reactors: G-L-S reactor, Slurry reactor, Bubble column reactor, Spray reactor, Trickle bed reactor	11	24
Unit– V	5.1 Performance equation of Single Ideal reactor for Single reaction Constant density system, Ideal batch	11	24

Design of single Ideal react	reactor, Steady state mixed flow reactor, Steady state plug flow reactor 5.2 Flow reactors: Space time, Space velocity, Holding time Vs. Space time		
	Total	**45**	**100**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
25%	**41%**	**34%**	---------	————————	————————

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S. No.	Title of Book	Author	Publication with place, year and ISBN
1	Chemical Reaction Engineering	Octave Levenspiel	Third Edition, John Wiley and Sons
2	Essentials of Chemical Reaction Engineering	H. Scott Fogler	Fourth Edition, Prentice Hall International
3	The Engineering of Chemical Reactions	Lanny D. Schmidt	Second Edition, Oxford University Press

(b) Open-source software and website:

Relevant NPTEL / SWAYAM online courses.

Suggested Practical List:

Sr. No.	Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology)	Approx. Hrs Required
1	To determine the activation energy of the reaction between Sodium thio Sulphate and HCl using Arrhenius Equation.	2
2	To determine order of the reaction between Sodium thio sulphate and HCl.	2
3	To determine the kinetics of a reaction between Ethyl Acetate and Sodium Hydroxide under condition of excess Ethyl Acetate at room temperature.	2
4	To determine the kinetics of a reaction between Ethyl Acetate and Sodium	2

	Hydroxide by integral method of analysis at room temperature.	
5	To determine the activation energy of the reaction between Ethyl Acetate and Sodium Hydroxide.	2
6	To determine the kinetics of a reaction between Ethyl Acetate and Sodium Hydroxide under by differential method of analysis at room temperature.	2
7	To determine the kinetics of a reaction between n- Butyl Acetate and Sodium Hydroxide by differential method of Analysis at room temperature.	2
8	To determine the kinetics of a reaction between n- Butyl Acetate and Sodium Hydroxide by integral method of Analysis at room temperature.	2
9	To determine the activation energy of the reaction between n- Butyl Acetate and Sodium Hydroxide using Arrhenius Equation.	2
10	To determine the rate constant and order of reaction for the catalytic decomposition of hydrogen peroxide.	2
11	To study the pseudo-first-order kinetics of the crystal violet–sodium hydroxide reaction and determine the activation energy.	2
12	Kinetic Study of Batch Reactor.	2
13	Kinetic Study of Continuous Stirred Tank Reactor.	2
14	Kinetic Study of Plug Flow Reactor.	2
15	Kinetic Study of Semi Batch Reactor.	2

Suggested Activities for Students:

1. Assignments
2. Technical Quiz/MCQ Test
3. Presentation on some course topic
4. I-net based assignments
5. Undertake micro-Project in team/individually

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